



Sri Lanka Institute of Information Technology

**B.Sc. Honours Degree in Information Technology
(Specialization: Data Science)**

**Final Examination
Year 3, Semester 1 (2019)**

**IT3021 – Data Warehousing and Business
Intelligence**

Duration: 2 Hours

Instructions to Candidates:

- ◆ This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence
- ◆ This paper has 5 questions. Answer all questions.
- ◆ Write answers in the booklet given.
- ◆ Total marks 100.
- ◆ This paper contains 6 pages including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1**(17 Marks)**

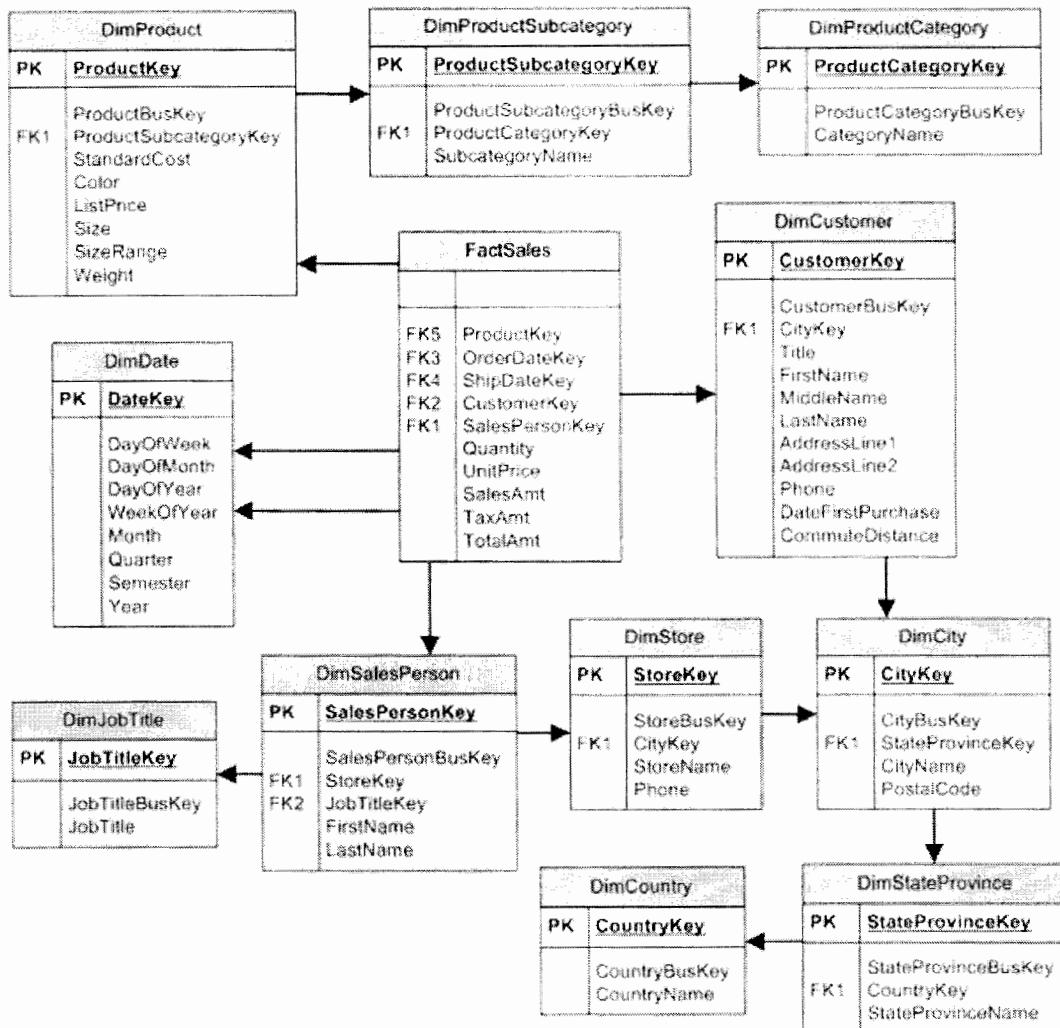
1. Explain what a Business Intelligence (BI) solution is with its key components. (3 marks)
2. Compare and contrast On-Line Transactional Processing (OLTP) systems with On-Line Analytical Processing (OLAP) systems in terms of their features. (4 marks)
3. Design a high-level solution architecture for a generic BI solution using at least five (5) main components of a BI solution. (5 marks)
4. A typical BI solution life cycle has different types of activities such as “*Launch Activities*”, “*Technology Track Activities*”, “*Data Track Activities*”, “*Application Track Activities*” and “*Wrap-up Activities*”.

List main activities of “*Technology Track*” and “*Data Track*”. (5 marks)

Question 2**(16 marks)**

1. Differentiate Fact tables and Dimension tables. (2 marks)
2. Explain the advantages of using surrogate keys in data warehouse modeling. (2 marks)
3. Compare and contrast Enterprise Data Warehouse (EDW) approach and Data Marts approach when designing a BI solution for an organization. (4 marks)
4. Model a Star schema for the given Snowflake schema design below.

State any assumptions you made when designing the model. (8 marks)



Question 3

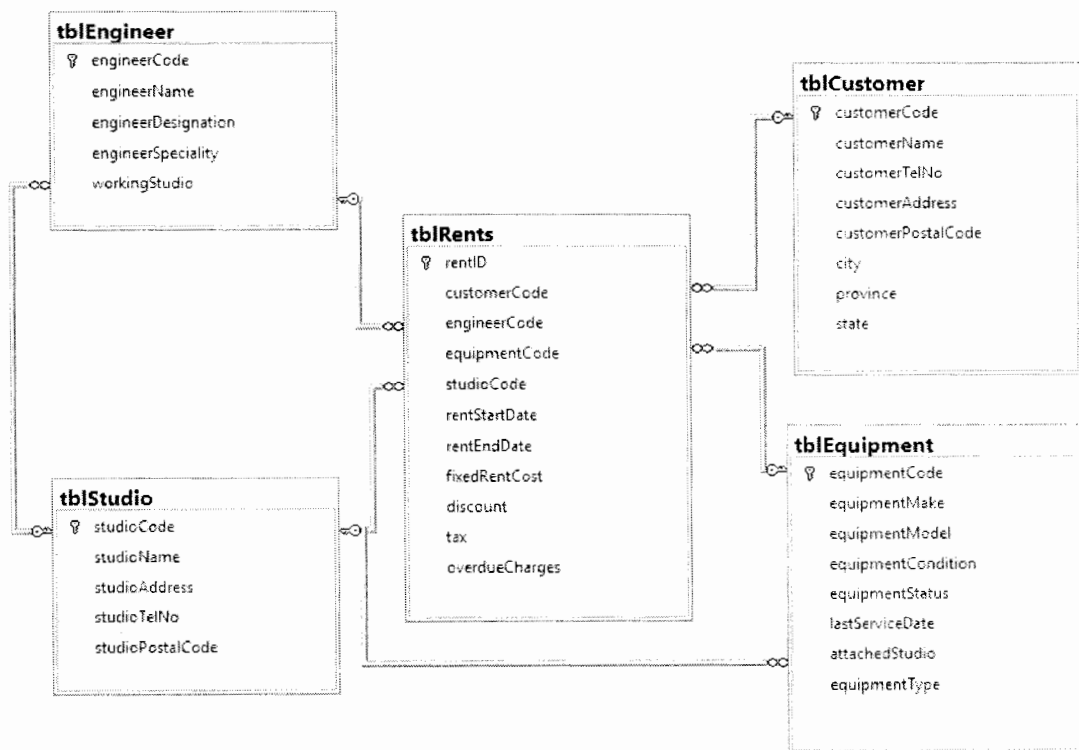
(30 marks)

- Consider the below ER diagram of an OLTP database. This is about a company which maintains studios in different locations. These studios rent out equipment it owns to their customers. Each studio is assigned with a set of engineers. Each equipment is attached to a particular studio. When an equipment is rented, an engineer attached to the particular studio is assigned for that rent session, so that the use of the equipment is overlooked by the engineer. Below is a brief description of their OLTP database.

Information:

- **tblStudio** table stores details of studio outlets including their locations.
 - o **studioCode** is the primary key of the table and is a business specific unique code
- **tblEquipment** table holds details of equipment rented out by the studio outlets of the company. Each equipment is attached to a studio

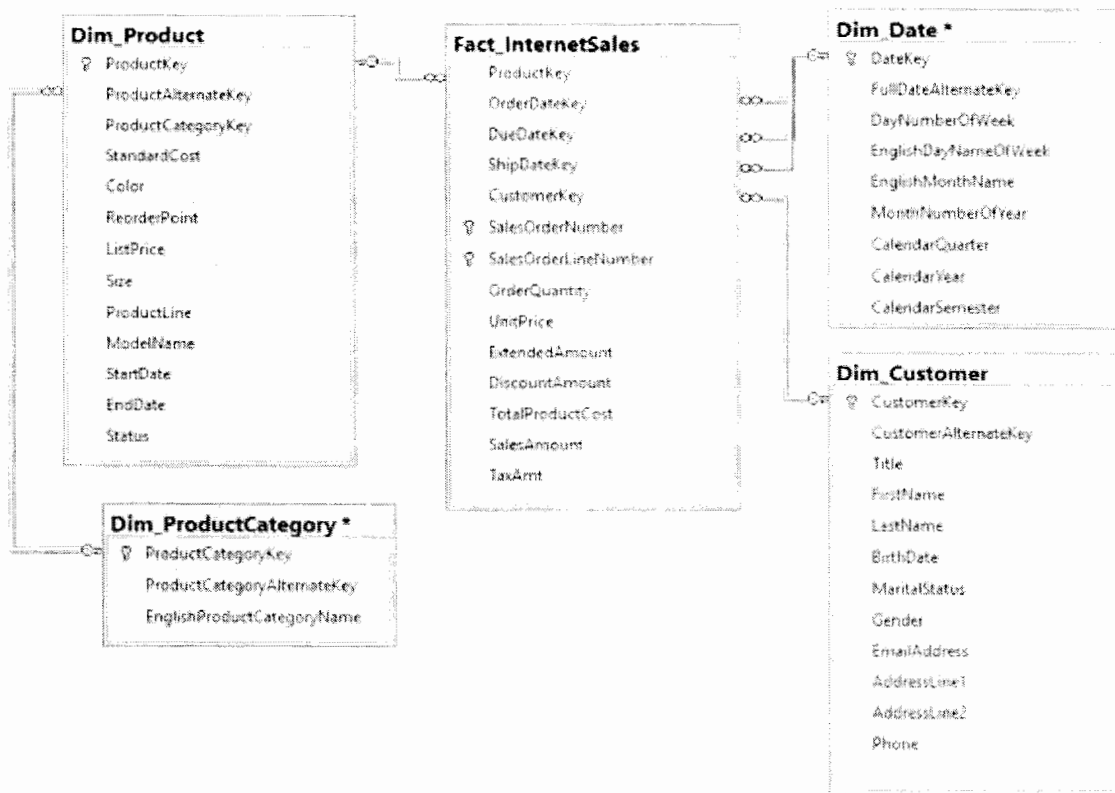
- *equipmentCode* is the primary key of the table and is a business specific unique code
- *attachedStudio* column is a foreign key to the *studioCode* of **tblStudio** table
- **tblCustomer** table holds the details about customers registered with the company. Customers may change their address information, and when they do it is updated in the system.
 - *customerCode* is the primary key of the table and is a business specific unique code
- **tblEngineer** table holds details of engineers. Each engineer is attached to studio.
 - *engineerCode* is the primary key of the table and is a business specific unique code
 - *workingStudio* column is a foreign key to the *studioCode* of **tblStudio** table
- **tblRents** table holds details of rental information.
 - *rentID* is the primary key of the table
 - *customerCode* column is a foreign key to the *customerCode* of **tblCustomer** table. This is to store to which customer the equipment is rented out for
 - *engineerCode* column is a foreign key to the *engineerCode* of **tblEngineer** table. This is to store which engineer is attached to overlook the equipment rented for this rent session
 - *equipmentCode* column is a foreign key to the *equipmentCode* of **tblEquipment** table. This is to store which equipment is rented for this rent session
 - *studioCode* column is a foreign key to the *studioCode* of **tblStudio** table. This is to store from which studio the equipment is rented



- a. Model a suitable dimensional model schema for a data warehouse for the given scenario (Star or Snowflake schema). State any assumptions you made for your model. Please note that the *workingStudio* column of **tblEngineer** table and *customerAddress* column of **tblCustomer** table should be maintained for historical data analysis.

(22 marks)

- b. Recommend two (2) hierarchies that can be derived in the data warehouse design? You may consider hierarchies which are not there in the OLTP system but can be derived in a DWH design for this scenario. (2 marks)
2. Consider the following Data Warehouse design. You need to model an ETL component to load data from source system (assume only one source system is available and all required data is extracted from the said source system) to the Data Warehouse. Note that the *Status* column of **Dim_Product** table should be maintained with historical data for changes over time.



When loading data to each of the above tables, state your recommendations considering the following aspects.

- Data loading sequence (order of tables in which data will be loaded)
 - Possible transformations (for the fact table and dimensions)
 - Possible data cleansing required (for dimensions)
- (6 marks)

Question 4**(18 marks)**

1. Differentiate OLAP storage types ROLAP, MOLAP and HOLAP in terms of following aspects:
 - a. How it is implemented
 - b. Advantages
 - c. Disadvantages(8 marks)
2. Explain following OLAP operations
 - a. Roll-up
 - b. Drill-down
 - c. Slice
 - d. Dice(4 marks)
3. Implementation of OLAP Cubes require additional investment since the implementation requires proprietary technologies. Assuming you are a BI solution architect that design & develops BI solutions for a company that generates millions of data daily and contains many departments, justify the requirement of investing in such a technology. (6 marks)

Question 5**(19 marks)**

1. Briefly explain what SSRS is. (2 marks)
2. Differentiate following tools in SSRS:
 - a. Report Builder
 - b. Mobile Report Publisher
 - c. Report Designer (SSDT)(6 marks)
3. Differentiate SSRS Drill-down reports and Drill-through reports. (4 marks)
4. Explain the use of KPIs available in SSRS (2 marks)
5. Explain the use of Subscriptions in paginated reports and state the available Subscription types in SSRS. (5 marks)